

DEAD BEAT CONTROL OF THREE PHASE PWM INVERTER

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ABSTRACT

This paper proposes new method of high performance current and/or voltage control of three phase PWM inverter, especially suitable for high speed switching devices such as Transistor and BIMOS.

As the controller for calculation of the dead beat algorithm, a high speed digital controller, such as the digital signal processor is assumed. Following new methods are successfully adopted and combined to realize high quality.

- (1) Three phase inverter current and voltage are transformed to synchronizing d-q-0 frame, and discretized equations of current and voltage are derived, which clearly reveal coupling of voltage and current of inverter on d-q frame.
- (2) Using above discretized equation of main circuit, algorithm of Dead Beat control with compensation for decoupling is developed.
- (3) New method for compensation of computing time delay and full utilization of sampling interval for inverter output pulse width is developed, which utilizes 2nd order load current prediction and discretization with twice time of sampling period.
- (4) Double Dead Beat control system is developed for sinusoidal output voltage inverter, which has high speed current minor loop to constrain inverter current within safety limit.

Validity of these techniques has been confirmed by simulation using a hybrid computer.

1. INTRODUCTION

Among many kinds of converters, the PWM inverter has most applications such as UPSs, active filters, high power factor converters and adjustable frequency drives.

Recently according to the progress of high speed switching devices and digital controllers such as DSP, new development stage of the PWM inverter can be expected.

Almost all applications of PWM inverters except UPS, have the controller structure with current minor loop, and performance of the inverter system largely depends on the quality of the current minor loop. Therefore current control of PWM inverter is one of the most important subject of power electronics.

In reference (1), the authors have proposed a new single phase BIMOS UPS with dead beat current minor loop and voltage major loop, implemented with two 8 bit micro-processors. And the authors

has presented high performance of current minor loop using high speed switching devices and dead beat control. And feasibility of UPS having current minor loop was also confirmed. Then it became possible to unify almost all application of PWM inverters including UPS with current minor loop.

Basing on above mentioned back ground, the authors have been trying to extend technology of dead beat current minor loop to three phase inverters. This paper present the principle of three phase sinusoidal voltage inverter with current minor loop, method of constructing dead beat current and voltage controller, and result of simulation using a hybrid computer.

2. PRINCIPLE OF SINUSOIDAL VOLTAGE INVERTER

Principle of proposed three phase sinusoidal inverter with current minor loop is shown in Fig.1. As the typical inverter main circuit, the three phase bridge inverter is supposed, but this principle can be used generally for other types of three phase voltage inverters, for example multi stage inverters using inter-phase reactors or multi-stage transformers.

To obtain high performance of control, the controller is constructed on synchronizing d-q frame. Three phase inverter variables such as V_C and I_C are transformed to variables on d-q frame such as $\hat{V}_C$ and $\hat{I}_C$. For simplicity PWM carrier is synchronized to sampling of digital control. When switching device can operate at higher switching frequency f_{sw} than the sampling frequency f_s , PWM frequency is selected to be $f_{sw} = n \cdot f_s$ ($n = \text{integer}$) as shown in Fig.2. This means that same pulse width is output two or three times in a sampling time. By doing so, inverter

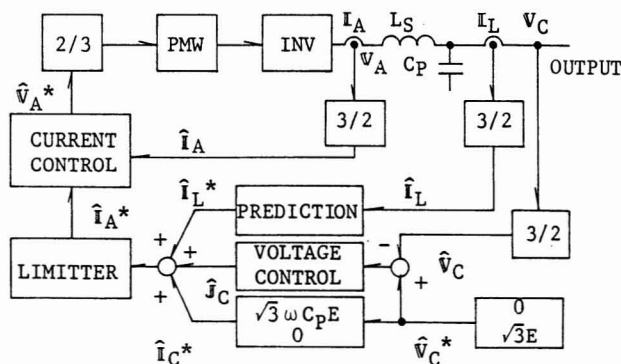


Fig.1 Sinusoidal voltage inverter
with current minor loop

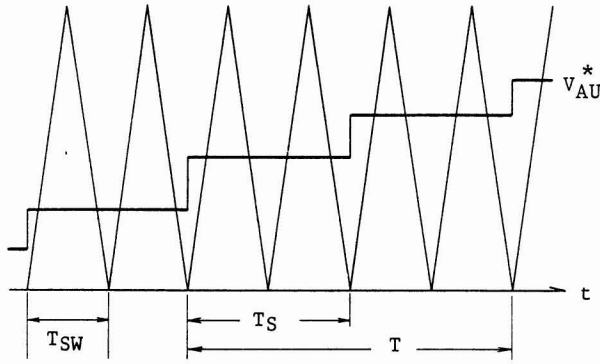


Fig.2 Relation of switching carrier, DDC signal and discretization time (when $n=2$)

can switch beyond audible frequency and ripple current can also be decreased, even if sampling frequency of DDC is lower than audible frequency.

Various methods of current control can be used for this system, but in this paper dead beat control is adopted as explained below. By using dead beat control, current response of one discretization time can be obtained.

As the reference of current minor loop, sum of three components $\hat{I}_L^*$, $\hat{I}_L^*$ and $\hat{I}_C$ are given through a limiter. No load voltage will be established by giving reference $I^* = \sqrt{3} \omega C_P E$ as current for parallel filter capacitance. On the one hand, load current $\hat{I}_L^*$ after discretization time T is given using prediction on d-q frame. The current minor loop follows this signal without delay and responses to distortion and disturbance of load.

The voltage controller stabilizes the system and assures constant voltage characteristic, by correcting instantaneous error of output voltage caused by following factors.

- Too fast change of load current that the current minor loop can not follow.
- Sudden change of di/dt of load current which causes error of prediction.
- Error of the current loop caused by deviation of circuit constant L_S , C_P from designed value.
- Error of the current loop caused by switching delay time T_d of the inverter.

If output voltage deviates by above factors, the load current changes from original wave form, so that the current loop follows this deviated load current through prediction, and results in unstable system. To stabilize the system against above mentioned factors, the voltage major loop also should have quick response. For that reason dead beat control of proportional type is adopted for the voltage major loop, and the authors call this system "double dead beat control system".

Providing a limiter for the reference of the current minor loop, easy applicable sinusoidal voltage inverter with inherent over current protection can be obtained. High speed compensation against d.c. voltage variation is performed when inverter voltage command $\hat{V}_A^*$ is transformed from d-q frame to 3 phase and given to PWM circuit.

3. DECOUPLED DOUBLE DEAD BEAT CONTROL SYSTEM

3.1 State equation of three phase expression

The main circuit of the three phase inverter is shown in Fig.3.

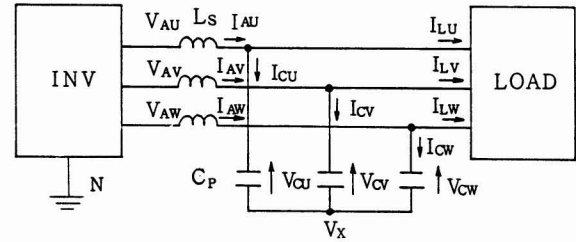


Fig.3 Main circuit

In this figure and following equations, meaning of variables and suffixes are as follows.

[Definition of variables]

- V_A ; inverter output terminal voltage from neutral
- V_C ; condenser voltage from V_X
- I_A ; inverter current
- I_C ; condenser current
- I_L ; load current
- V_X ; condenser neutral voltage from neutral
- U, V, W ; components of three phase expression
- d, q ; components of d-q frame expression
- $*$; command value
- ω ; angular velocity of output frequency
- T ; discretization time
- $\wedge$; vector expression of variables on q-d-0 or q-d frame

Using vector equation, main circuit of Fig.3 can be expressed by following equations

$$C_P \frac{d}{dt} V_C = I_A - I_L \quad \text{----- (1)}$$

$$L_S \frac{d}{dt} I_A = V_A - V_C - V_X \quad \text{----- (2)}$$

where each variables such as I_A , V_C and etc, are vector variables given by equations (3) with suffixes of each variable. L_S and C_P are given by equations (4), and V_X is given by following equation (5).

$$I = \text{col} [I_u, I_v, I_w] \quad \text{----- (3)}$$

$$V = \text{col} [V_u, V_v, V_w] \quad \text{----- (4)}$$

$$L_S = \text{diag} [L_s, L_s, L_s] \quad \text{----- (5)}$$

$$C_P = \text{diag} [C_p, C_p, C_p] \quad \text{----- (6)}$$

$$V_X = (V_{AU} + V_{AV} + V_{AW}) / 3 \quad \text{----- (7)}$$

From equation (2) and (5)

$$L_S \frac{d}{dt} I_A = D V_A - V_C \quad \text{----- (8)}$$

where

$$D = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \quad \text{----- (9)}$$

3.2 State equation on synchronizing d-q frame

Equation (1) and (6) will be transformed to q-d-0 frame by using following transformation matrix C , and relation of (9).

$$C = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos \omega t & \cos(\omega t - 2\pi/3) & \cos(\omega t + 2\pi/3) \\ \sin \omega t & \sin(\omega t - 2\pi/3) & \sin(\omega t + 2\pi/3) \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \quad (8)$$

$$\hat{V} = \begin{bmatrix} V_q \\ V_d \\ V_0 \end{bmatrix} = C \begin{bmatrix} V_u \\ V_v \\ V_w \end{bmatrix} = C V \quad \text{----- (9)}$$

$$\hat{I} = \begin{bmatrix} I_q \\ I_d \\ I_0 \end{bmatrix} = C \begin{bmatrix} I_u \\ I_v \\ I_w \end{bmatrix} = C I$$

As the result of transformation, equation (6) reduces to equation (10).

$$\begin{bmatrix} 0 & \omega L_s & 0 \\ -\omega L_s & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \hat{I}_A + L_s \frac{d}{dt} \hat{I}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \hat{V}_A - \hat{V}_c \quad \text{--- (10)}$$

Since 0-axis components has no relation to other axes, these components can be omitted from equation (10).

Then state equations (11) of inverter current can be derived.

$$\begin{bmatrix} \dot{I}_{Aq} \\ \dot{I}_{Ad} \end{bmatrix} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \begin{bmatrix} I_{Aq} \\ I_{Ad} \end{bmatrix} + \begin{bmatrix} 1/L_s & 0 \\ 0 & 1/L_s \end{bmatrix} \begin{bmatrix} V_{Aq} \\ V_{Ad} \end{bmatrix} - \begin{bmatrix} 1/L_s & 0 \\ 0 & 1/L_s \end{bmatrix} \begin{bmatrix} V_{cq} \\ V_{cd} \end{bmatrix} \quad \text{----- (11)}$$

Next, as the result of transformation of condenser voltage equation (1), following equation (12) can be derived.

$$\begin{bmatrix} 0 & \omega C_p & 0 \\ -\omega C_p & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \hat{V}_c + C_p \frac{d}{dt} \hat{V}_c = \hat{I}_A - \hat{I}_L \quad \text{----- (12)}$$

Also in this equation, 0-axis components can be omitted, and state equation (13) of condenser voltage can be derived.

$$\begin{bmatrix} \dot{V}_{cq} \\ \dot{V}_{cd} \end{bmatrix} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \begin{bmatrix} V_{cq} \\ V_{cd} \end{bmatrix} + \begin{bmatrix} 1/C_p & 0 \\ 0 & 1/C_p \end{bmatrix} \begin{bmatrix} I_{Aq} \\ I_{Ad} \end{bmatrix} - \begin{bmatrix} 1/C_p & 0 \\ 0 & 1/C_p \end{bmatrix} \begin{bmatrix} I_{Lq} \\ I_{Ld} \end{bmatrix} \quad \text{----- (13)}$$

Reference of output voltage and condenser current on synchronizing frame will be given as follows.

First, output voltage reference of equation (14) is transformed by matrix C as equation (15), and reference value $\sqrt{3}E$ on d-axis is obtained.

$$V_c^* = \begin{bmatrix} V_{cu}^* \\ V_{cv}^* \\ V_{cw}^* \end{bmatrix} = \begin{bmatrix} \sqrt{2} E \sin(\omega t) \\ \sqrt{2} E \sin(\omega t - 2\pi/3) \\ \sqrt{2} E \sin(\omega t + 2\pi/3) \end{bmatrix} \quad \text{--- (14)}$$

$$\hat{V}_c^* = \begin{bmatrix} V_{cq}^* \\ V_{cd}^* \end{bmatrix} = C V_c^* = \begin{bmatrix} 0 \\ \sqrt{3} E \end{bmatrix} \quad \text{----- (15)}$$

Equation (16) of condenser current reference by three phase expression is transformed as equation (17).

$$I_c^* = C_p \frac{d}{dt} V_c^* \quad \text{----- (16)}$$

$$\begin{aligned} \hat{I}_c^* &= \begin{bmatrix} I_{cq}^* \\ I_{dq}^* \end{bmatrix} = C I_c^* = C C_p \frac{d}{dt} V_c^* = C C_p \frac{d}{dt} (C^{-1} \hat{V}_c^*) \\ &= \left[C C_p \frac{d}{dt} C^{-1} \right] \hat{V}_c^* + C C_p C^{-1} \frac{d}{dt} \hat{V}_c^* \\ &= \begin{bmatrix} 0 & \omega C_p \\ -\omega C_p & 0 \end{bmatrix} \hat{V}_c^* + \begin{bmatrix} \sqrt{3} \omega C_p E \\ 0 \end{bmatrix} \quad \text{----- (17)} \end{aligned}$$

3.3 Discretization of the state equations

Here state equation (11) and (13) of inverter current and voltage will be discretized separately. Separate two step discretization of current equation (11) and voltage equation (13) means

(i) Condenser voltage V_c is assumed to be constant during discretization time T of inverter current.

(ii) Inverter is assumed to be ideal current source and its current is assumed to be constant during discretization time T of condenser voltage.

This method is simple and has sufficient accuracy when $\sin \omega T \approx \omega T$ is valid. Also discretized current equation and current minor loop of this method can be applied widely for arbitrary type of converter configurations other than fig.1 which has no condenser C_p .

First, equation (11) is rewritten as equation (18), and matrix A and F of its discretized state equation (19) are derived.

$$\dot{\hat{I}}_A(t) = A' \hat{I}_A(t) + F' \hat{V}_A(t) - F' \hat{V}_c(t) \quad \text{----- (18)}$$

$$\hat{I}_A(k+1) = A \hat{I}_A(k) + F \hat{V}_A(k) - F \hat{V}_c(k) \quad \text{----- (19)}$$

Next, equation (13) is rewritten as equation (20), and matrix K of its discretized form (21) is derived.

$$\dot{\hat{V}}_c(t) = A' \hat{V}_c(t) + K' \hat{I}_A(t) - K' \hat{I}_L(t) \quad \text{----- (20)}$$

$$\hat{V}_c(k+1) = A \hat{V}_c(k) + K \hat{I}_A(k) - K \hat{I}_L(k) \quad \text{----- (21)}$$

Method and result of above mentioned matrix calculation is as follows. (5)

$$A = \mathcal{L}^{-1} [(sI - A')^{-1}] = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$F = \int_0^T A (T - \tau) F' d\tau = \begin{bmatrix} c & -d \\ d & c \end{bmatrix} \quad \text{----- (22)}$$

$$K = \int_0^T A (T - \tau) K' d\tau = \begin{bmatrix} e & -f \\ f & e \end{bmatrix}$$

$$\left. \begin{aligned} a &= \cos \omega T, \quad b = \sin \omega T, \quad c = \frac{1}{\omega L_s} \sin \omega T, \\ d &= \frac{2}{\omega L_s} \sin^2 \frac{\omega T}{2}, \quad e = \frac{1}{\omega C_p} \sin \omega T, \\ f &= \frac{2}{\omega C_p} \sin^2 \frac{\omega T}{2} \end{aligned} \right\} \quad \text{----- (23)}$$

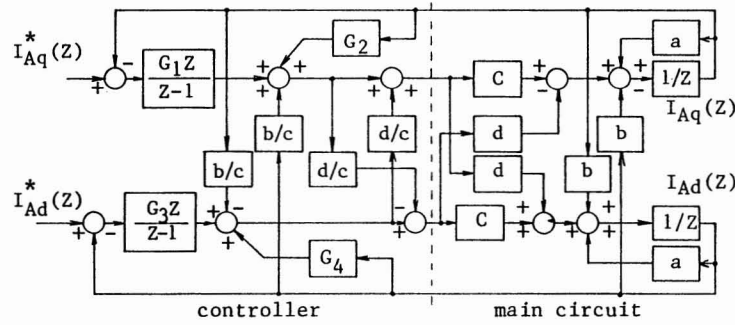


Fig.4 Decoupled dead beat current control system

Once inverter operating frequency ω , circuit constant L_s , C_p and discretization time T are determined, above elements of matrices become constants.

3.4 Decoupled dead beat current controller

Applied voltage $\hat{V}_{AC}$ to the reactor L_s is adopted as the control variable.

$$\hat{V}_{AC}(k) = \hat{V}_A(k) - \hat{V}_C(k) \quad \text{----- (24)}$$

And equation (24) is applied to equation (19), and the resultant equation is Z transformed as

$$Z \hat{I}_A(Z) = A \hat{I}_A(Z) + F \hat{V}_{AC}(Z) \quad \text{----- (25)}$$

This equation can be expressed by a block diagram as shown in the right side of the dotted line of fig.4

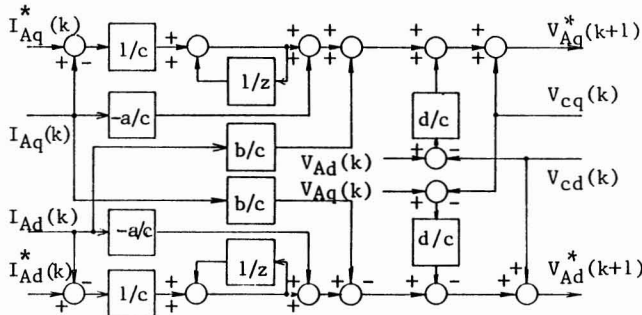


Fig.5 Current controller

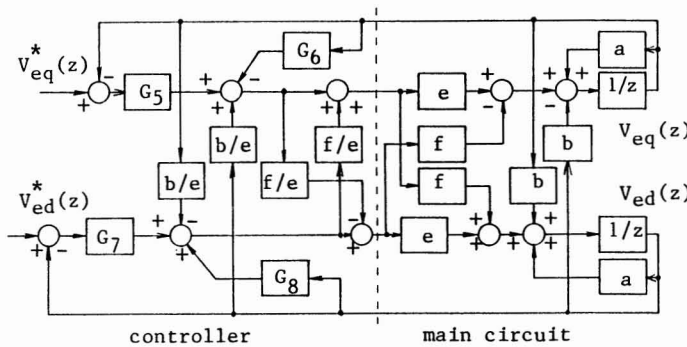


Fig.6 Decoupled dead beat voltage control system

This block diagram clearly shows the discretized relation of current and voltage of three phase inverter in d-q frame.

Couplings of d-q components by terms "b" and "d" are decoupled by feed-forward terms "b/c" and "d/c".

After removing coupling as mentioned above, then dead beat controllers of I-P function is provided as shown in the left side of the dotted line of fig.4.

Adopting I-P control, influence of switching dead time T_d to avoid short through can be compensated quickly when inverter current crosses zero.

To get finite time settlement response, condition to nullify two roots of loop transfer function of block diagram of fig.4 is derives as

$$G_1 = G_3 = 1/c, G_2 = G_4 = -a/c \quad \text{----- (26)}$$

Adding the d-q components of V_C , the current controller of Fig.5 can be obtained.

3.5 Decoupled dead beat voltage controller

To get the equation of relation between error $\hat{V}_E$ of output voltage and output $\hat{J}_C$ of the voltage controller, following modification of equation will be performed.

If inverter is assumed to become a ideal current source by the current minor loop of the paragraph 3.4, its current can be expressed as

$$\hat{I}_A = \hat{I}_C^* + \hat{I}_L^* + \hat{J}_C \quad \text{----- (27)}$$

Error of condenser voltage is set as

$$\hat{V}_E(k) = \hat{V}_C(k) - \hat{V}_C^*(k-1) \quad \text{----- (28)}$$

Reference of the condenser voltage has following relation with reference of condenser current.

$$\hat{V}_C^*(k) = A \hat{V}_C^*(k-1) + K \hat{I}_C^*(k) \quad \text{----- (29)}$$

Using equation (27), (28) and (29), following equation (30) can be derived from equation (21).

$$\hat{V}_E(k+1) = A \hat{V}_E(k) + K \hat{J}_C(k) \quad \text{----- (30)}$$

Z transform of this equation is

$$Z \hat{V}_E(Z) = A \hat{V}_E(Z) + K \hat{J}_C(Z) \quad \text{----- (31)}$$

This equation (31) can be expressed by a block diagram shown in the right side of the dotted line of fig.6. This block handles only the error of the condenser voltage, so reference for each component is of course zero.

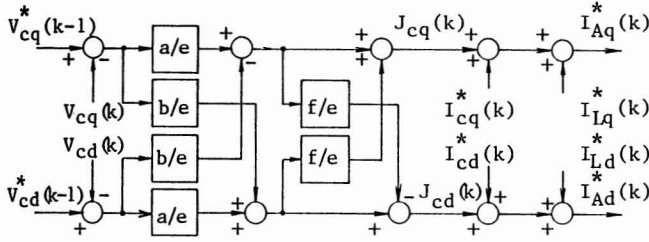


Fig.7 Voltage controller

After providing feed-forward terms "b/e" and "f/e" to cancel the coupling terms "b" and "f", then dead beat controllers of P operation is provided for each component as shown in the left side of the fig.6.

Condition for finite time settlement response of this voltage controller is

$$G_5 = G_7 = 1/e, G_6 = G_8 = (1-a)/e \quad \text{---- (32)}$$

Changing the signal from error voltage to condenser voltage and adding I_{C*} and I_{L*} , voltage controller of Fig.7 can be obtained.

3.6 Simplification of controller

Coefficients of the controllers obtained in paragraph 3.4 and 3.5 can be given by following equations using equation (23).

$$\left. \begin{aligned} a/c &= \omega L_S \cot \omega T & \approx L_S/T &= 1.95 \\ b/c &= \omega L_S & &= 0.189 \\ 1/c &= \omega L_S / \sin \omega T & \approx L_S/T &= 1.95 \\ a/e &= \omega C_p \cot \omega T & \approx C_p/T &= 0.78 \\ b/e &= \omega C_p & &= 0.075 \\ d/c &= f/e = \tan \omega T/2 \approx \omega T/2 & &= 0.048 \end{aligned} \right\} \text{--- (33)}$$

where example of actual coefficients are obtained for typical circuit constants of simulation in chapter 5. That is $\omega = 2\pi f = 377$ rad/sec, $T = 128 \times 2$ micro sec, $L_S = 500$ microH and $C_p = 200$ microF. When ωT is sufficiently small as in this case, above approximation is valid. Then using this simplified coefficients, total controller

obtained as the combination of Fig.5 and 7 becomes as shown in Fig.8. By above approximation, number of terms which change according to frequency, decreases, then this controller can be easily applied for adjustable frequency inverters. And also if $\omega T/2$ is sufficiently small, this term can be eliminated.

4. PREDICTION OF LOAD CURRENT AND COMPENSATION OF COMPUTING DELAY

In case of dead beat control, controller output during a discretization time is determined definitely if the target value at the end of this discretization time is given.

The target value, that is the reference for the current minor loop of fig.1 is the sum of $\hat{I}_C^*$ and $\hat{I}_L^*$.

Condenser current command $\hat{I}_C^*$ at the end of each discretization time and $\hat{I}_C$ can be given internally, then to get the target value $\hat{I}_L^*$ of load current, the second order prediction method (3) is successfully adopted.

Another important problem is compensation of computing time delay and full utilization of inverter pulse width to enable output of inverter voltage while the processor is executing computation of this discretization time.

For this purpose, discretization time T of twice the sampling time T_S , that is $T=2T_S$, is adopted. In chapter 3, "k" is used as the step of discretization time, then to distinguish from "k", "m" will be used as the step number of sampling.

Referring to fig.9, if time t is mT_S now, target value of load current $I_L^*(m)$ at $t_S = (m+2)T_S$ is predicted using $I_L(m-2)$, $I_L(m-1)$, $I_L(m)$ according to second order prediction formula of (34).

$$I_L^*(m) = 6 I_L(m) - 8 I_L(m-1) + 3 I_L(m-2) \quad \text{---- (34)}$$

To realize load current $I_L^*(m)$ shown by a mark 0 in fig.9, after the discretization time $T=2T_S$, computing time T_1 shown in fig.9 is necessary for the processor, so result $V_A(m)$ of the former sampling time is given to the PWM circuit during this sampling time $mT_S - (m+1)T_S$.

Then if the result of the computation is $V_{A0}(m)$ as shown by the line X_1-X_2 in fig.9, so area S_A

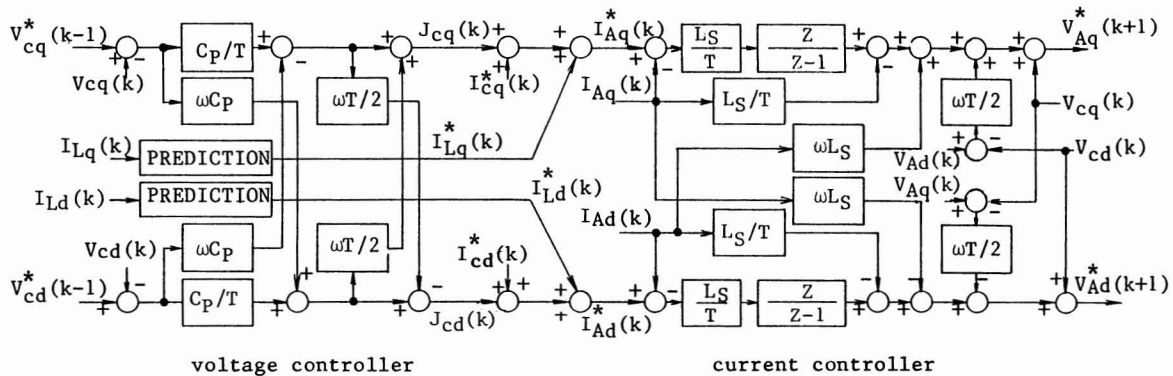
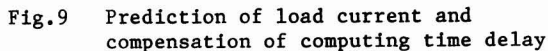


Fig.8 Simplified voltage and current controller


$$V_A(m+1) = 2V_{A0}(m) - V_A(m) \quad \text{-----} (35)$$

Another current target value J_C to correct voltage error is supplied from voltage controller which operates, as can be seen from equation (28), on the difference between voltage reference of one discretization time before $V_C^*(m-2)$, and the condenser voltage at present $V_{C(m)}$.

Using a high speed DSP, this sampling time can be shortened to around 100 to 200 micro seconds, so high speed current and voltage control by software which is suitable for high speed switching device such as BIMOS can be realized.

Characteristics of the system concerning following parameters have been studied to confirm validity of this principle.

- Beside usual wave forms, circular vector loci of inverter voltage and current were adopted to get better understanding of inverter operation. Typical results of simulation will be introduced below.

Throughout following simulations, circuit condition of CASE A₁ is mostly used. This condition corresponds to BIMOS inverters. In some case, condition of CASE A₂ which suppose a very high speed controller is used.

[CASE A,

[CASE A₂]

L_S and C_P ; ditto, $T_S = 80$ micro sec, $f_{sw} = 12.5$ kHz (one triangular wave per T_C)

[CASE B]

$L_S = 2.5\text{mH}$, $C_P = 500$ micro F star, $T_S = 520$ micro sec, $f_{sw} = 1.92$ kHz (one triangular wave per T_S)

(1) Influence of sampling interval T_s and switching delay T_d

Simulation wave forms and vector loci for linear series L-R load are shown in Fig.10.

Distortion of the output voltage V_C slightly increases when T_d of 3 micro sec is assumed for BIMOS inverter, as shown in Fig.10-1 and 10-2. However, if the sampling interval T_S is shortened from 128 micro sec to 80 micro sec, influence of T_d is suppressed very rapidly and output voltage distortion becomes very small as shown in Fig.10-3.

Signal V_A^* , which is the command value to PWM circuit, shows operation of the digital controller to correct influence of T_d .

In case of transistor inverter, the filter reactor L_S and the capacitor C_p are enlarged to cope with long sampling interval ($T_d = 30$ micro sec). Then even in this case, fairly good sinusoidal output voltage V_C can be obtained as shows in Fig.10-4 and 10-5.

(2) Tracking characteristics to load harmonics

To examine influence of load harmonics, a ideal three phase current source load, which is composed of 100% fundamental component and 20% or 10% harmonic component, is connected as a load.

Result for 7th harmonic shown in Fig.11, indicate excellent response of this system. But in case of 13th harmonic, some tracking error

Fig.10 Vector loci of
inverter current and
voltage
(Linear L-R load 9KVA)

Fig.10-1
Case A₁
 $T_S = 128\mu S$
 $T_d = 0$

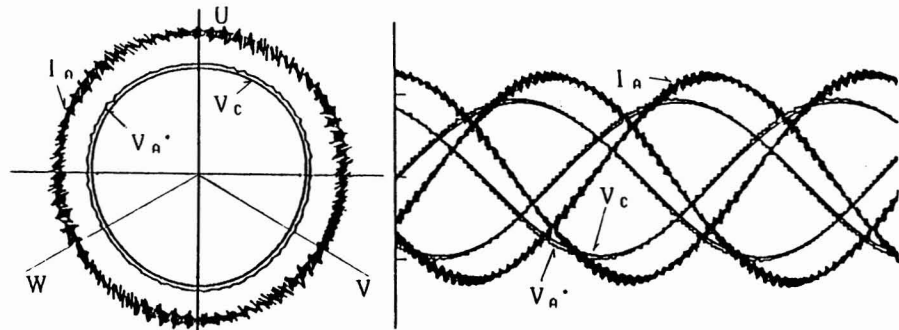


Fig.10-2
Case A₁
 $T_S = 128\mu S$
 $T_d = 3\mu S$

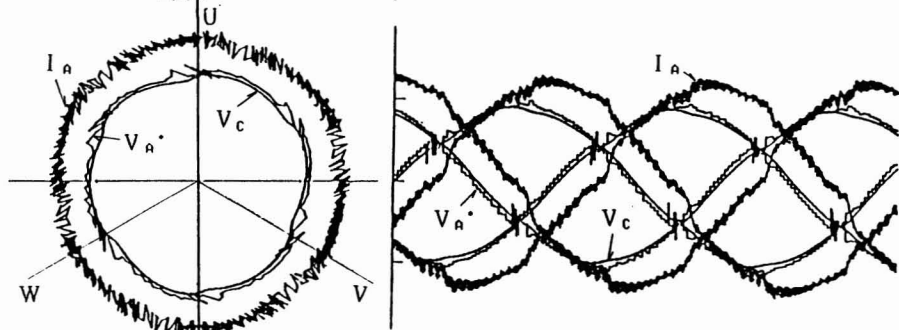


Fig.10-3
Case A₂
 $T_S = 80\mu S$
 $T_d = 3\mu S$

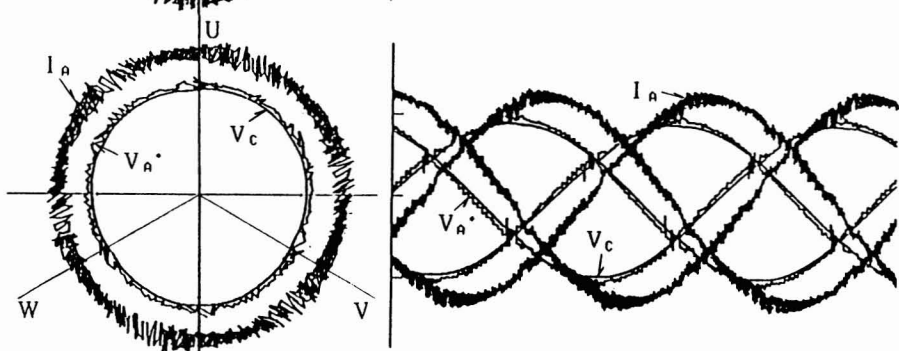


Fig.10-4
Case B
 $T_S = 520\mu S$
 $T_d = 0$

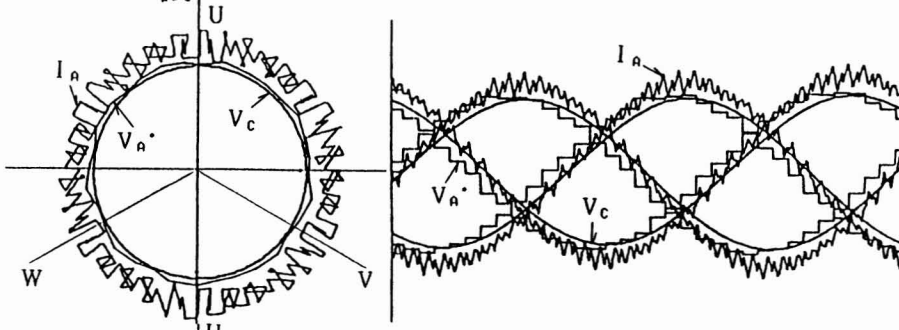
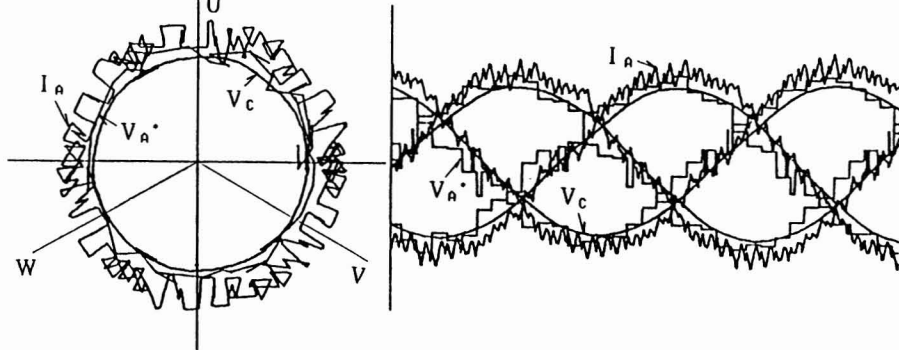


Fig.10-5
Case B
 $T_S = 520\mu S$
 $T_d = 30\mu S$



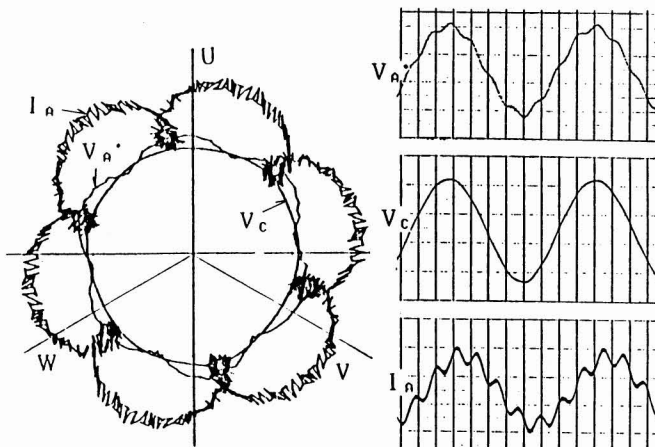


Fig.11
Vector loci of inverter
current response for load
current of 7th harmonics
(Condition: Case A₁, T_d = 0
I_L; fundamental = 30A
7th = 6A, 20%)

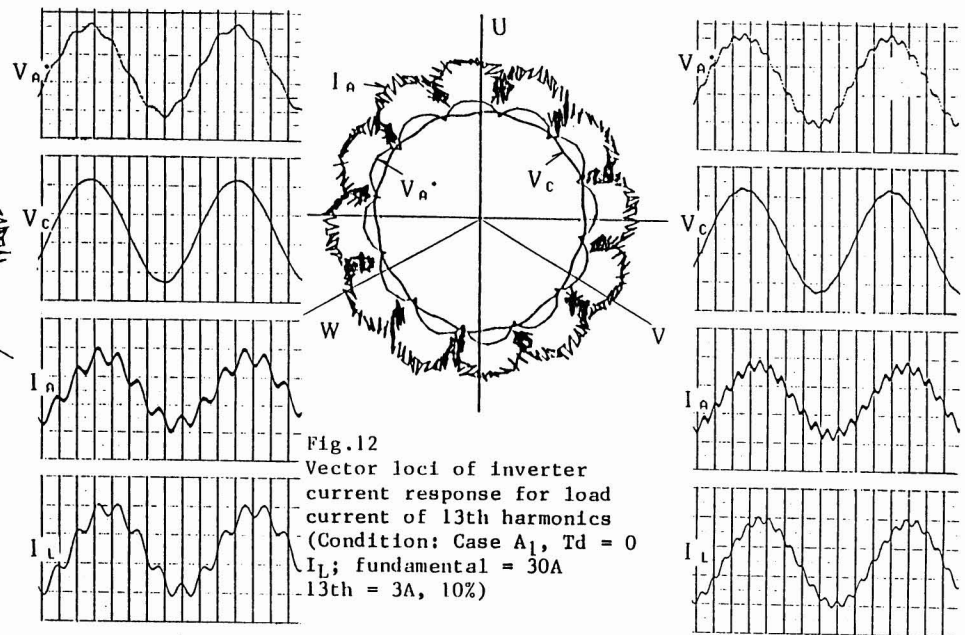


Fig.12
Vector loci of inverter
current response for load
current of 13th harmonics
(Condition: Case A₁, T_d = 0
I_L; fundamental = 30A
13th = 3A, 10%)

Fig.13
Vector loci of inverter
current and voltage for
adjustable frequency
operation
(Condition: Case A₁, T_d = 0
series L-R load, 0 ~ 60 Hz)

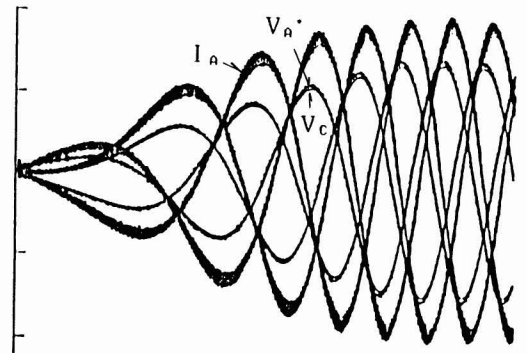
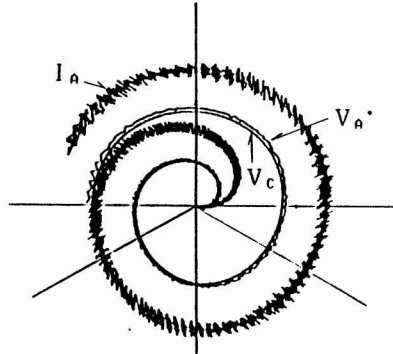
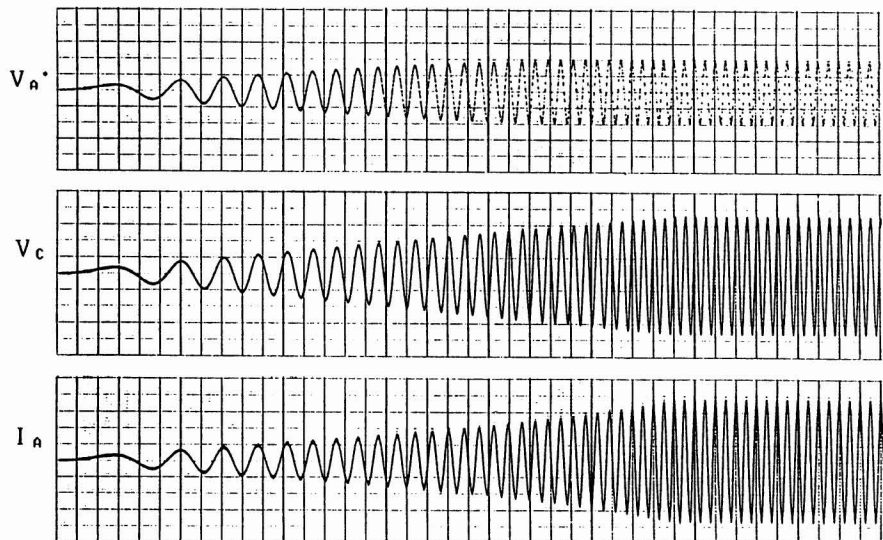


Fig.14
Output waveform under
adjustable frequency and
voltage operation
(Condition: Case A₁, T_d = 0
series L-R load, 0 ~ 400 Hz)



occurs, because sampling interval T_s of 128 micro sec correspond to 36 degree for 13th harmonics, therefore error of 2nd order prediction and delay of control bring a little deterioration of current tracking.

(3) Adjustable frequency operation

This system can be easily adopted as an adjustable frequency and voltage inverter for a.c. motor drive. In this case, the synchronizing rotating frame for transformation and reference of output voltage and condenser current should be changed according to frequency.

Simulation result of operation from 0 to 60 Hz is shown in Fig.13.

Another example of higher output frequency is shown in Fig.14

6. CONCLUSION

To construct dead beat control system of three phase inverter, relation of voltage and current components of three phase inverter was first clarified in discretized form on d-q frame. Then algorism of dead beat current minor loop and voltage major loop for sinusoidal voltage inverter was developed.

New method of compensation of computing time delay which enables full utilization of PWM period for output voltage, was also developed.

Validity of this control system was proved by simulation using a hybrid computer.

Principle of the current minor loop proposed in this paper can be applied commonly to various applications of PWM inverters.

Another useful result of this paper is introduction of equation (25) and block diagram of Fig.4 which precisely takes coupling term $\omega T/2$ into consideration. This equation can be used for design of conventional digital P-I current controller of PWM inverter or cycloconverter, and can be expected to bring better result.

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